

Other than the increase in die space, what would be the impact of increasing the number of registers?

10. What is the difference between in-order and out-of-order processors? What kind of processor is Nehalem? What kind of processor is the Power 7 processor?

Algorithmic questions

1. If you are given the choice to select between a greedy programming solution and a dynamic programming solution to a problem, which would you select? Justify your choice.
2. What is the lower bound on the complexity of any sorting algorithm that uses only comparisons to make its decisions? Can this be improved by using additional knowledge on the kind of inputs that are processed?
3. Given a directed graph which contains negative cycles, is it possible to find the shortest paths between any two pairs of vertices? Justify your answer.
4. You are given an array of N integers. Consider this array as a set of N integers (there are no duplicates in the array). Consider the subsets of this set whose cardinality is $(N-1)$. How many such subsets are there? There are $(N-1)$ such subsets. For each such subset, your program should output the product of the elements present in it. For instance, if you are given the array containing 10, 20, 30, 40, we have the 3 subsets, {10,20,30}, {20,30,40}, and {30,40,10}. The algorithm should output the 3 values 6000, 24000 and 12000. Can you come up with an $O(N)$ algorithm for the same?
5. Given the problem of finding out whether a vertex with the specified label is present in a given sparse undirected graph, which grows

progressively fat as the depth from the root increases, which one of the two graph-traversal algorithms (namely, breadth-first search or depth-first search) would you use? Justify your answer.

My 'must read book' for this month

This month's book suggestion comes from Vivek K Rajan. His recommendation is '*The Algorithm Design Manual*' by Steve Skiena. While this book is not meant as a first course in algorithms (I would recommend that you read '*Introduction to Algorithms*' by Cormen first), it is a great find for those trying to sharpen their skills in algorithm design. It also contains 'war stories', which describe the author's practical experience in trying to come up with an algorithm to solve a real problem. The insights one can obtain from reading these war stories go a long way in helping to understand the programming and design choices that each programmer needs to make in solving a given problem. Do make time to read this book.

If you have a favourite programming book that you think is a "must-read book for every programmer", please do send me a note about your recommendation, along with a short description on why you think it is useful. I will feature your choice in this column. This would help many readers who want to improve their coding skills.

Please do send me your answers to this month's questions. If you have any favourite programming puzzles you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming, and here's wishing you the very best! 

About Sandya Mannarswamy

The author is a specialist in compiler optimisation and works at Hewlett-Packard India. She has a number of publications and patents to her credit. Her areas of interest include virtualisation technologies and software development tools. If you are preparing for computer science programming interviews, you may find it useful to visit Sandya's programming interviews discussion group 'Computer Science Interview Training (India)' on LinkedIn.

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